

RESEARCH ARTICLE

Developing a Transparent Diagnosis Model for Diabetic Retinopathy Using Explainable AI

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This work was supported by the Research Fund from the University of Johannesburg, Johannesburg, South Africa.

ABSTRACT Diabetic retinopathy is a leading cause of vision complications and partially sighted which pose considerable diagnostic difficulties because of its diverse and varying symptoms. Some of them include the fact that the disease displays a non-uniform pattern, where patients present different symptoms; the requirement of highly qualified specialists to interpret the images of the fundus; the risk of errors in the interpretation of images or their inconsistency; and the absence of clear morphological signs often makes early diagnosis unlikely. Traditional diagnosis mostly rely on the expert interpretation of retinal images, which can lead to bias and inaccuracy; highlighting the need for improved diagnostic methods. Although traditional Artificial Intelligence (AI) methods enhance the diagnostic capabilities remarkably, their black box nature and information opacity restrict healthcare providers to comprehend the reasoning framework of the AI to build trust and optimize its usage in practice. Explainable AI (XAI) is an emerging approach that addresses the black-box problem by improving the interpretability of models, which allows users to understand the logic behind certain decisions. This research proposed a diagnosis model for detecting diabetic retinopathy using XAI approaches that increases the interpretability of the models to help clinicians understand the reasons behind the decisions. The proposed model is used to enhance diagnostic accuracy, offer comprehensible, and concise insights regarding the diagnostics. The convergence history plots of the proposed model validate the learning process to achieve 94% better diagnostic accuracy than traditional methods while improving interpretability and applicability in healthcare settings, indicating improvement in accuracy and loss reduction.

INDEX TERMS Diabetic retinopathy (DR), artificial intelligence (AI), explainable AI (XAI), convolutional neural network (CNN).

I. INTRODUCTION

Diabetic Retinopathy (DR) is an ocular disease that affects the retina in the back of eye by damaging its blood vessels. This is a complication that results from impaired blood circulation in the retina due to damage in the small blood vessels which

The associate editor coordinating the review of this manuscript and approving it for publication was Wei Wang¹.

leads to their leakage and swelling and formation of abnormal new blood vessels in its most severe form. These changes can affect vision and in severe cases cause partially sighted if not well addressed. Diabetic retinopathy cannot be prevented but the development and progression can be slowed by frequent eye examinations & better control of blood sugar [1], [2].

DR is one of the major causes of visual impairment that can be prevented and is common among adults of 20 to

74 years [3], [4]. Screening for DR must be done frequently to identify advanced complications such as proliferative diabetic retinopathy and diabetic macular edema that cause irreversible vision loss. Screening for DR in the primary care has been proved to be effective in decreasing the risk of visual loss, especially if the patients also have optimized diabetes management [5]. Deep learning based on convolutional neural networks has been shown to achieve high sensitivity and specificity in diagnosing the final stages of DR; thus, they are useful for screening large populations in primary care practices [6]. Therefore, there is a need to design interventions to tackle problems like financial barrier, transport challenge, and no PCP in order to enhance the DR screening in this group of people mainly those who are preferably of low income and from the underrepresented groups.

The prevalence of diabetes mellitus has increased around the world, which has heightened the need for proper approaches toward the treatment of DR [7], [8]. DR is ranked as one of the most prevalent diabetes-related complications, and if left uncontrolled, it results in considerable vision loss. The crucial aspect of preventing such outcomes is the early identification of the diabetic retinopathy; however, this is a challenge [9]. The disease develops gradually, often asymptomatic in the initial phases, which hampers early diagnosis of the condition. Eye care management for diabetes is highly relevant, simple regular screening, and better diagnostic techniques for identifying DR at an early stage, allows one to act and avoid vision loss among diabetic patients [10], [11].

It is crucial to have an accurate depiction of diabetic retinopathy since its onset cannot be overemphasized. The early diagnosis of symptoms and accurate diagnosis of the disease make it easier to treat the condition with little or no threat to one's vision. The routine diagnosing of DR has historically relied on the ophthalmoscopic inspection of fundus photographs usually by an ophthalmologist. Although these are useful approaches, they are lengthy, cumbersome, noisy and carry the extra tag of variability due to human interference. This underscores the necessity of employing automated and accurate diagnostic systems [12], [13].

Recent developments in AI have brought new prospects for the assessment of DR [14]. Deep learning models have shown high degree of accuracy in diagnosing retinal images for different purposes such as in predicting risk factors of cardiovascular disease (CVD) [15], for the diagnosis of ophthalmic diseases [16], and for detecting age-related macular degeneration (AMD) [17]. However, these models have some limitations especially due to the fact that they are opaque and the processes by which they make decisions are difficult to understand by healthcare professionals and patients [18].

Interpretability and understanding, explainability, are considered as major goals of AI since black box models pose a great question [19]. XAI is used to enhance the transparency of systems as well as make AI decision-making more transparent and credible [21]. XAI produces human understandable explanations that can assist the user in comprehending how and why a specific decision was made, which

helps in building credibility. This transparency is beneficial not only in terms of proper feature recognition and model verification but also when addressing such important issues as controlling AI bias and practicing responsible AI usage.

This study combines XAI with deep learning models as a way of identifying diabetic retinopathy, offering clear and transparent details of the prediction factors, thus making the AI system accurate and reliable. The objective is to convey the message that incorporating explainability into AI systems can bridge the gap between technological innovation and clinical practice for better patient prognosis in DR.

II. LITERATURE REVIEW

Diabetic retinopathy is a complication of diabetes and the leading cause of partially sighted among the working population worldwide. In order to avoid the more serious vision loss the detection and treatment of DR should occur in the early stages. Traditional diagnostic approaches entail the use of physical examination by ophthalmologists; these can be tiresome and prone to inter-observer differences. The integration of AI has brought about potential innovations in diagnosing the DR through use of approaches for example deep learning to improve the accuracy of diagnosis. However, many AI models are a black box, which has caused some concern with regards to interpretability and transparency. There are limited studies that analyze the use of AI techniques in diagnosing DR today, which deal with developing diagnostics that would help make AI-driven diagnostics more effective for practitioners.

This research work [21], asserted that there has been slow adoption of artificial intelligence and data analysis in DR screening compared to the progress in other developments such as laser surgery and drug service aimed at managing the risks associated with diabetes. It demonstrates the importance of the machine learning model in the diagnosis of DR using Linear Support Vector Classifier (LinearSVC), Calibrated ClassifierCV, and Logistic Regression model which was an 80% predictive value of the disease. Through the utilization of data collected in the National Health and Nutrition Examination Survey, the study explained how decision-making has transitioned towards the use of data. Specifically, this research focused on gender, insulin, hypertension, and duration of diabetes in predicting DR, and the shown process was proved to be one of multiple factors.

In this research [22], the authors examined the potential of essential factors like, HbA1c, disease duration, and albumin excretion rate in diabetic patients to gauge the probability of DR. This focused on the associations between different risks and diabetes complications and these associations are established to be useful in creating prediction models. This employed curve fitting methods such as the Artificial Neural Networks (ANN) and the least squares regression as enhanced by the software applications Tiberius and the Statistical Package for the Social Services (SPSS). Seven distinct regression patterns: linear, logarithmic, quadratic, cubic, power, S-curve, and exponential approaches were

considered in building the prediction models. The analysis further showed that the higher R-squared figures depict better regression models than both ANN and other regression types.

In [23], the authors have directed their attention toward the use of deep learning or machine learning algorithms to extract important information related to risk factors to generate early warning of DR. It presented a new approach involved deep neural network (DNN) and Recursive Feature Elimination (RFE) for traditional risk factor base early DR diagnosis improvement. The proposed model significantly improved the accuracy of predicting DR in its initial stages, achieving a prediction accuracy of 82.033%, surpassing current best-practice models. The study highlighted on extraction of necessary risk factors employing RFE and stress was laid down on the significance of diagnosis of retinopathy relying on these factors. This research work also affirmed the stability and transferability of the proposed prediction model by comparing other machine learning models and bringing nephropathy and hypertension-diabetes datasets in light of predicting other diabetic complications.

In this research [24], the use of ML algorithms in DR prediction has been well illustrated with excellent accuracy indicators including sensitivity, specificity, percentage of true positive, false negative rate, and receiver operating characteristic (ROC) graphs. This research used different datasets and classified different ML algorithms for analyzing datasets and to decide the effective models for solving problems such as Naive Bayes, Sequential Minimal Optimization (SMO), logistic regression, Stochastic Gradient Descent (SGD), bagging classifier, J48 classifier, decision tree classifier, and random forest classifier. Specifically, the results have showed that the technique of logistic regression has a higher average accuracy among the mentioned methods, which proves its high potential as a tool for promoting early diagnosis and prevention of DR. This highlighted the powerful contribution of the ML in increasing the accuracy and efficiency of DR screening and diagnosis.

This research [25] focused on one of the major complications of diabetes known as retinopathy which if not detected can lead to partially sighted. It highlighted how data mining could be instrumental in early identification of diabetic retinopathy, leading to a better healthcare system. The research demonstrated the possibilities of using classification models and some of the machine learning algorithms with hyperparameters optimization that allowed achieving satisfactory accuracy level of diabetic retinopathy. Diabetes is associated with life-threatening conditions such as cardiovascular disease, stroke, nephropathy, diabetic foot, and retinopathy; hence, it requires proper identification and control. Machine learning algorithms with the selected hyperparameters demonstrated greater accuracy in improving prediction models, thus proving valuable for public health boons. This focused on utilizing such advanced machine learning methodologies to enhance the diagnosis and early detection of Diabetic Retinopathy.

The study [26] aimed at using machine learning and severity indices for mortality in diabetic ICU patients, which highlights the importance of diabetes, care, and complications on ICU outcomes. Prior research has pointed diabetes as one of the critical health risks that leads to renal, cardiac, and neuropathic problems, among other ones that increases the healthcare costs, ICU admissions, and death rates. It was distinguished using Medical Information Mart for Intensive Care III (MIMIC-III) database and by using machine learning algorithms combined with binomial logistic regression for developing the models for risk of mortality in diabetic patients in the ICU. The efficiency of the models was pleasing with AUC values showing good fit at 0.787 and 0.785, as seen from the analysis confirming the reliability of the predictions made by both models. The main variables included HbA1c, mean glucose levels during the stay in the ICU, admission diagnoses, age of the patient and type of admission and this was more efficient than other models requiring more variables for the same predictions.

This research work [27] looked at the increasing concern of diabetes as a chronic disease that has affected millions of people around the world; with many others going undiagnosed in the asymptomatic stage. Engaging data from patients in Sylhet Diabetes Hospital in Bangladesh, this work also emphasized the use of real-world data for analysis and forecasting purposes. Data mining methods such as Naive Bayes, Logistic Regression, and Random Forest were adopted and assessed using ten-fold Cross Validation and the Percentage Split method to determine the best approach for predicting risky diabetes. The research was concluded by the design of a widget for diabetes risk estimation based on symptoms with additional tips for the management of risk factors, focusing on the utility of the study.

III. LIMITATIONS OF THE PREVIOUS APPROACH AND OUR CONTRIBUTIONS

Several approaches have been proposed to ensure accurate diagnosis and analysis in the context of healthcare data communication. However, these typical mechanisms are less appropriate for DR detection systems because medical data have high variability and complexity. Several research works have explored the problems of accuracy and interpretability of DR in various contexts, and the results reflect thoughtful gaps in the modern approach that requires further development. Specifically, DR detection systems grapple with challenges such as:

A. LACK OF ROBUST PREPROCESSING LAYERS

The efficiency of preprocessing layer directly determines the improvement of the image quality, ensure data consistency, as well as improve feature extraction. If there is no preprocessing, models face poor image quality, resulting in low accuracy and lead to reduced accuracy and generalizability.

B. LIMITED USE OF EXPLAINABLE AI (XAI)

In this context, XAI refers to the ability to understand and interpret the model's decisions. The lack of such XAI techniques can hinder the identification of errors, as well as the clinicians' trust in the AI model.

Table 1 indicates the limitations of previous approaches such as limited data preprocessing [25], [26], [27], [32], [34] the trust issue, low transparency, and lower accuracy [28], [29], [30], [31], [32], [33], [34].

By addressing these challenges, the proposed research is incorporating preprocessing layer for image enhancement, and adopting XAI method to develop more reliable, interpretable, and clinically applicable models for the diagnosis and grading of diabetic retinopathy.

IV. PROPOSED MODEL

In recent times, diagnosing diabetic retinopathy has faced numerous hurdles due to the disease's diverse manifestations and often elusive early symptoms. The complexity of interpreting retinal images, combined with indistinct early indicators of the disease, makes the diagnostic process highly susceptible to human error and variability. Traditional diagnostic methods leads to inconsistent result and AI leads to black-box nature that hinders understanding, confidence, and practical application in clinical settings. Explainable AI (XAI) offers a solution with its white-box nature by making the AI's decision-making process transparent and comprehensible. This research proposed an XAI-based model to explain the decision made by the AI to the healthcare professionals hence building their confidence when using the AI in diagnosing the diabetic retinopathy to improve the patients' care. The proposed is model is depicted in figure 1 below.

As shown in the figure 1, the first step of the proposed model is obtaining retinal images from patients. These images are used for the assessment of diabetic retinopathy as well as for prevention of worse vision deterioration. After obtaining the retinal images, a rigorous preprocessing layer is applied to the data in order to prepare it for effective model training. This phase entails data augmentation which refers to the process of manipulating and enhancing the collected images so that the size and heterogeneity of the dataset is boosted. After augmentation, the data is split into three distinct sets: with a ratio of training (60%), testing (20%), and validation (20%). It helps in training, testing, and validation of the model where each set is used for a unique process in the model. Then the split training set containing 60% of the data undergoes a deep learning based convolutional neural network (CNN) algorithm that is best suited to learning and detecting patterns of DR. This phase is an iterative process since the CNN fine-tunes its parameters in order to produce the least erroneous predictions. This proposed algorithm implicates that the retinal image variations can be complicated and sophisticated, however, this model can predict the outcomes successfully. These predictions are then taken to XAI methods to explain the rationale of the decisions to make them

transparent (white-box), to explain the features and patterns of the retinal images used in the diagnosis.

The primary goal of XAI is to build trust in deep learning (black-box) by elucidating the importance of local and global variables through post hoc explanations. By unravelling the complexities of black-box models, XAI promotes accountable AI and helps in developing transparent models. Enabling explainability in deep learning for diabetic retinopathy diagnosis makes it easier for clinicians and stakeholders to understand the rationale behind algorithmic decisions. Then, it is checked, if the learning rate is met, the trained explainable patterns are stored in the cloud; otherwise, the model retrain the deep learning based CNN algorithm.

In the validation phase, the trained model stored in the cloud and the stored data for validation is imported for prediction. The validation data is used to check the efficacy of a model, which is less sensitive to errors. Further, the testing data is utilized to evaluate the model's performance, providing a final assessment of its effectiveness before deployment in real-world scenarios. The proposed model then checks for the diabetic retinopathy or its absence. If the evaluation algorithm indicates diabetic retinopathy, a message is provided stating that the disease has been identified. Otherwise, the process is terminated and no more steps are made towards the process and material.

V. SIMULATION RESULTS

To overcome these critical challenges for diagnosing diabetic retinopathy at the appropriate stage, extensive testing and validation of the proposed XAI-based model is needed. This section describes the rigorous testing and validation procedure for demonstrating the viability, accuracy, and robustness of the model. The process of developing the model is applied to a dataset [35], comprised of a training phase in which 60% (1702 samples) of the data set is used to train the model to recognize certain features that are characteristic of diabetic retinopathy, a validation phase where 20% (568 samples) of the data set is used to validate the hyperparameters of the model and determine whether the model generalizes well from the provided training data and to avoid the problem of overfitting, a testing phase which uses the remaining 20% (568 samples). Thus, the primary goal of the proposed research is to show that the proposed model is feasible and can be directly applied in actual clinical contexts after a careful validation and fine-tuning of the model.

As shown in the figure 2, it presents the class distribution of the training dataset for diagnosing DR. The x-axis represents the two classes: The presence of Diabetic Retinopathy is coded as DR while the absence of Diabetic Retinopathy is coded as No_DR. The y-axis measures the number of samples contained within each class. The bars are colored blue for "DR" and green for "No_DR" to differentiate the two groups. This bar chart can also show the number of samples in each class when training the model so that the organizer can determine the balance of the training dataset.

TABLE 1. Comparative analysis of proposed model with previous published works.

Authors	Challenges	Method	Preprocessing Layer	Decision-Making	XAI	Limitations
Baiju et al., 2023 [25]	Early diagnosis of diabetic retinopathy to prevent vision loss; improving prediction accuracy using data mining and machine learning techniques.	Machine Learning Techniques	×	✓	×	Faces limitations like data quality, model complexity, overfitting risk, computational demands, lack of real-world validation, biases, integration challenges, and a narrow focus on one condition
Anand et al., 2018 [26]	Predicting mortality in diabetic ICU patients; addressing the impact of diabetes, health maintenance, and comorbidities on ICU outcomes.	Machine learning techniques and binomial logistic regression to create mortality risk prediction models.	×	✓	×	focused on specific variables such as HbA1c, glucose levels, admission diagnoses, patient age, and type of admission.
Islam et al., 2020 [27]	Addressing the undiagnosed cases of diabetes due to its asymptomatic phase; improving prediction accuracy using real-world data.	Naive Bayes, Logistic Regression, and Random Forest	×	✓	×	Limited by data quality, model interpretability, overfitting risk, computational demands, real-world validation, patient data biases, and clinical integration challenges.
Sarwinda et al., 2018 [28]	Classifying stages of Diabetic Retinopathy (DR) due to image similarity across stages.	Support Vector Machines (SVM) and Random Forest (RF)	✓	✓	×	Relatively small dataset (111 images); need for larger dataset for further validation; varying performance across different DR stages.
Khojasteh et al., 2018 [29]	Identifying diabetic retinopathy signs (exudates, hemorrhages, microaneurysms) with improved accuracy; enhancing CNN performance with preprocessing techniques.	Convolutional Neural Network (CNN)	✓	✓	×	Relatively small dataset (89 images); need for larger datasets to further validate results; variation in performance across different DR signs.
Dutta et al., 2018 [30]	Identifying key antecedents of Diabetic Retinopathy (DR) using deep learning models with higher accuracy.	backpropagation NN, Deep Neural Network (DNN), and Convolutional Neural Network (CNN); Fuzzy C-means algorithm	✓	✓	×	Requires larger datasets for validation; preprocessing steps impact feature loss and accuracy; variation in image resolutions and noise.
Qummar et al., 2019 [31]	Detecting different stages of Diabetic Retinopathy (DR) with high accuracy; addressing low performance in early stages.	Ensemble of five deep CNN models (Resnet50, Inceptionv3, Xception, Dense121, Dense169);	✓	✓	×	Small dataset size; need for larger datasets to further validate results; varying performance across different DR stages, especially the mild stage.
Bhardwaj et al., 2021 [32]	Automatic detection and grading of diabetic retinopathy (DR) using transfer learning and deep learning models; addressing the need for early-stage detection to prevent severe blindness.	convolution neural networks (CNNs) with statistical feature selection module and SVM classifier	×	✓	×	Requires validation on larger datasets; high computational complexity; varying performance across different CNN models.
Nazih et al., 2023 [33]	Detecting the severity stages of diabetic retinopathy (DR) using vision transformer (ViT) models; addressing data imbalance.	CNN models like ResNet50, InceptionV3, and VGG19.	✓	✓	×	Requires validation on larger datasets; high computational complexity; varying performance with different hyperparameters and data balancing techniques.
Atwany et al., 2022 [34]	Manual DR detection is time-consuming and prone to intra-grader inconsistencies; early detection in less prominent stages.	Deep learning methods	×	✓	×	Varied dataset quality; imbalance in data distribution; need for standardized evaluation metrics.

A balanced dataset helps the model to learn how to classify the two conditions properly while an imbalanced the model is likely to overemphasize the frequent class.

The figure 3 intends to capture the random sample images from the training set used for DR diagnosis and the annotations as ‘DR’ for the images with the disease and ‘No_DR’ for the images without the disease. It is useful for explaining the

nature and variety of images in the dataset and understanding the data that is inputted into the diagnostic model.

The figure 4 shows the mean images for two classes: “DR” (Diabetic Retinopathy) and “No_DR” (No Diabetic Retinopathy). The left subplot displays the average image calculated from all samples in the training dataset labeled as “DR,” while the right subplot shows the average image

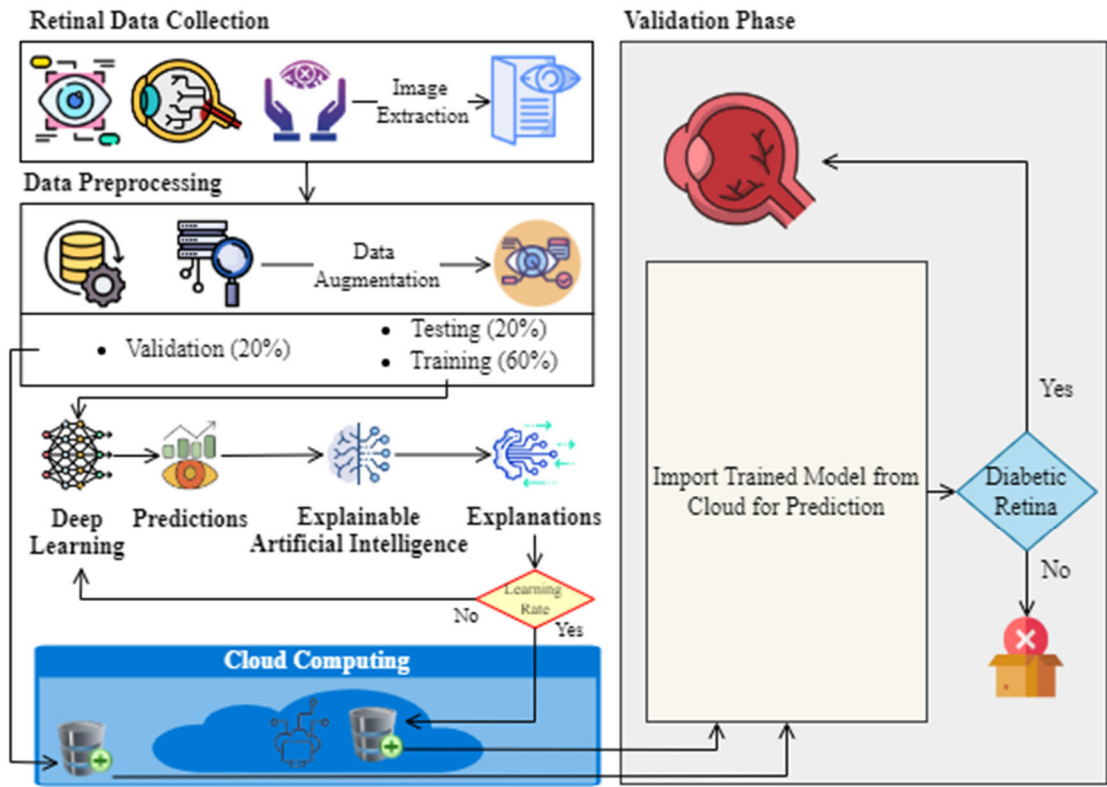


FIGURE 1. Proposed model for diagnosing diabetic retinopathy.

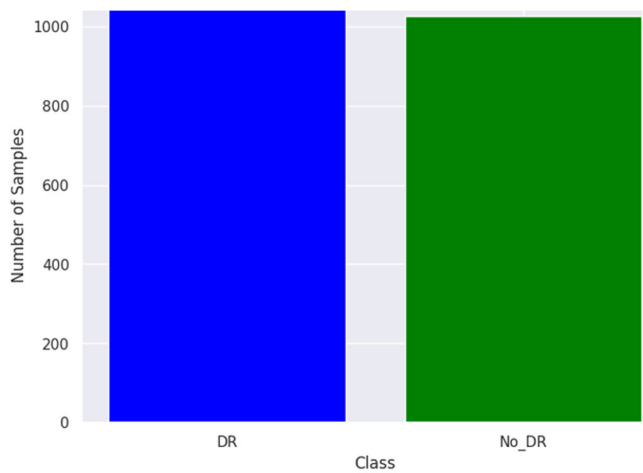


FIGURE 2. Classification distribution in training dataset.

from samples labeled as “No_DR.” These mean images are created by averaging the pixel values of all images within each class, providing a visual representation of the typical characteristics for each class. This is helpful in determining the similarities that are shared amongst the images associated with diabetic retinopathy compared with those not associated with the disease and hence see the distinguishing features that the AI model will look for when learning.

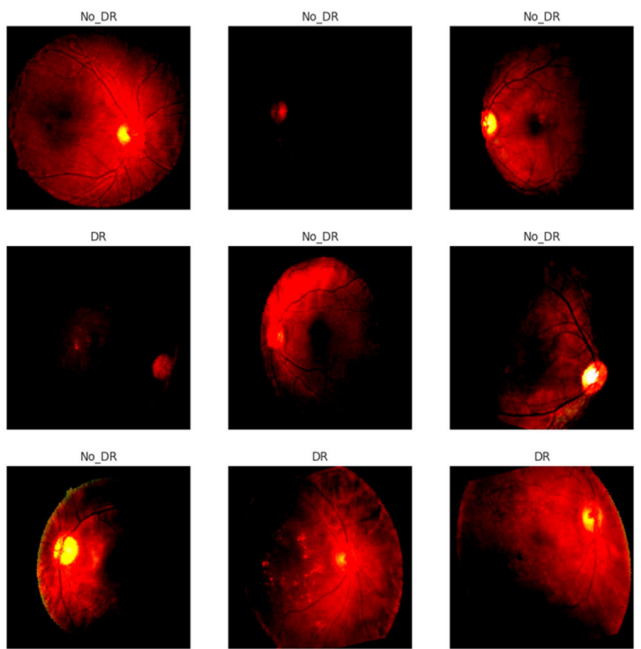


FIGURE 3. Random sample images with notations.

Figure 5 is representing a random sample image from the training dataset used for diagnosing diabetic retinopathy (DR) along with its corresponding histogram of pixel

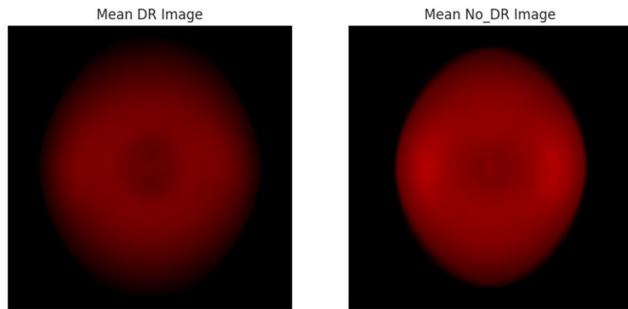


FIGURE 4. Mean images of each class.

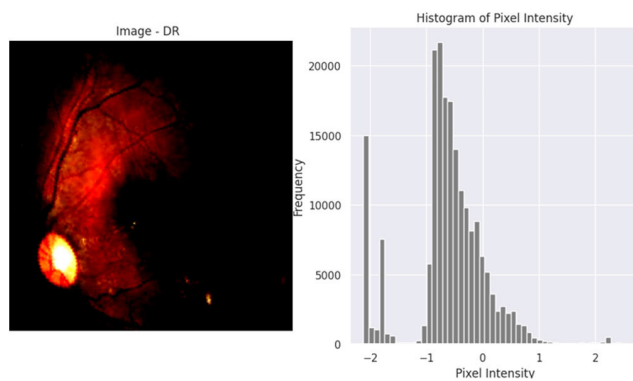


FIGURE 5. Image with overlaid histogram of pixel intensity.

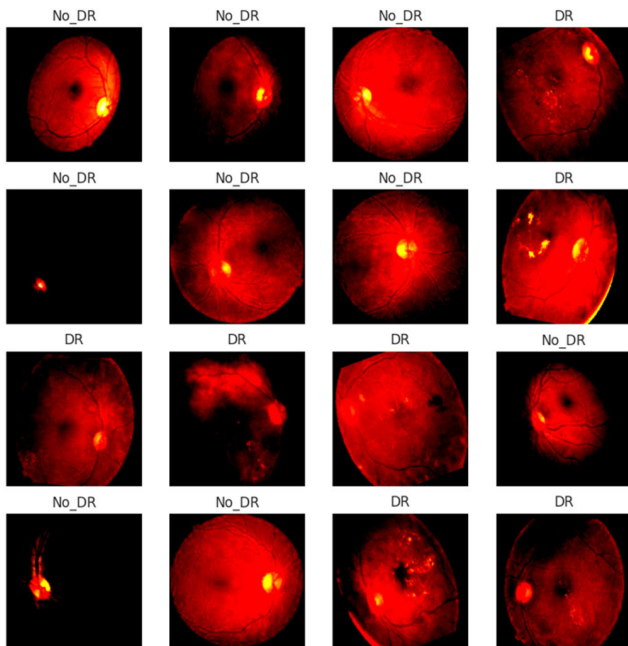


FIGURE 6. Retinopathy images.

intensity. The left side of the figure displays the image with a title indicating its class label. The right side shows a histogram that plots the pixel intensity distribution of the image, where the x-axis represents the range of pixel intensities, and the y-axis shows the frequency of each intensity level.

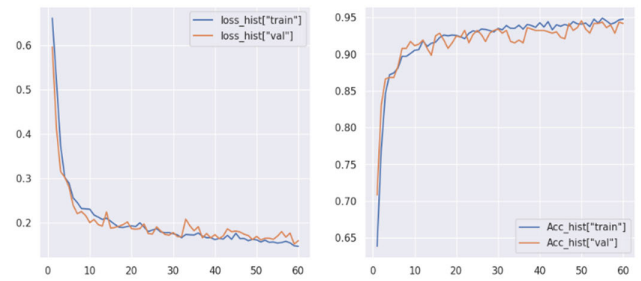


FIGURE 7. Convergence history of the model over training epochs.

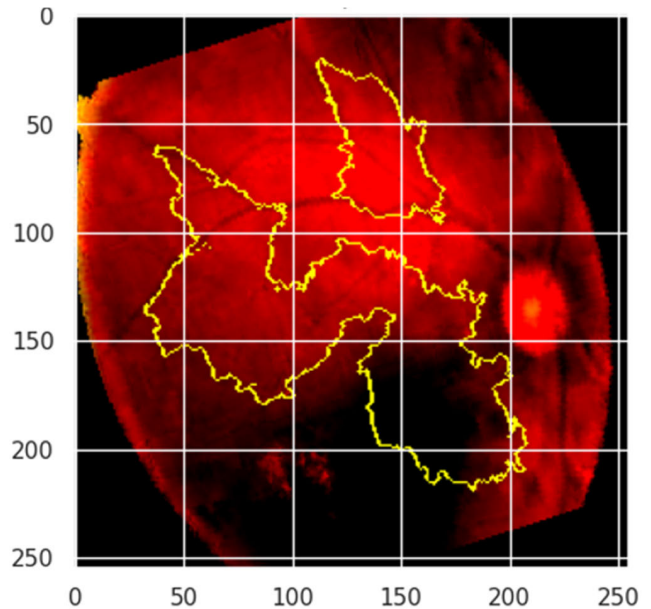


FIGURE 8. LIME explanation.

This figure provides a comprehensive view of both the visual characteristics of the sample image and the statistical distribution of its pixel intensities, aiding in the understanding of the quality and features of the image data used in training the diagnostic model.

The figure 6 is showing a 4×4 grid of retinal images from the training dataset used for classifying diabetic retinopathy. Each image is labeled either 'DR' (Diabetic Retinopathy) or 'No_DR' (No Diabetic Retinopathy), providing a visual overview of the dataset's content. This visualization also provides an indication of the nature of the images which are displayed and aids in the comparison of healthy and diseased retinæ. The displayed images allow for quick inspection of the dataset, aiding in assessing the quality and variability of the data, which is crucial for developing effective machine learning models for diagnosing diabetic retinopathy.

The figure 7 illustrates the convergence history of the model over 50 epochs, showcasing two subplots: the left plot shows the training and validation loss while the right plot shows the training and validation accuracy. In the left plot, the training loss (blue line) starts at 0.8 and decreases to

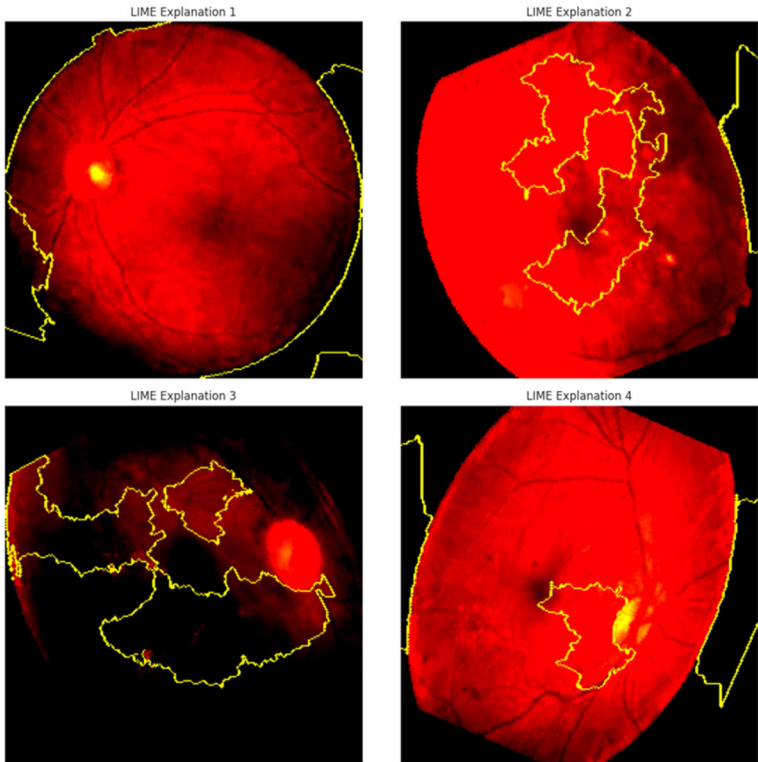


FIGURE 9. LIME explanation for 4 test images.

0.1, while the validation loss (orange line) starts at 0.85 and reduces to 0.15, indicating effective learning and improved performance with minor fluctuations around the 30th epoch. In the right plot, the training accuracy (blue line) begins at 40% and rises to 95%, whereas the validation accuracy (red line) starts at 45% and increases to 94%, reflecting enhanced accuracy and generalization capabilities. Altogether, these plots show how the model’s learning process reduces loss and increases accuracy over epochs, as well as generalizing from the training data to the validation data.

The figure 8 and 9 is showing the Local Interpretable Model-agnostic Explanations (LIME) explanations for the model’s predictions on four selected images from the test set. Each subplot visualizes one test image with highlighted boundaries, indicating the most influential regions that contributed to the model’s decision-making process. The boundaries highlight the top features that had the greatest impact on the classification, providing a clear and interpretable visual explanation of the model’s reasoning for each image. The evaluation of LIME explanations for various test images provides an opportunity to gain a better understanding of the model’s workings and improve confidence in its decisions. The figure is emphasizing its focus on providing insights into the model’s decision-making for multiple cases.

Table 2 demonstrates a performance comparison of the proposed model using XAI with previous approaches [28], [29], [30], [31], [32], [33], [34]. It is indicated that the pro-

TABLE 2. Comparative analysis of the proposed model as compared to the previously published approaches.

Authors	Method	Accuracy (%)	Miss-rate (%)
Sarwinda et al., 2018 [28]	Support Vector Machines (SVM) and Random Forest (RF)	85.23	14.77
Khojasteh et al., 2018 [29]	Convolutional Neural Network (CNN)	87.6	12.4
Dutta et al., 2018 [30]	Backpropagation NN, Deep Neural Network (DNN), and Convolutional Neural Network (CNN)	BNN=62.7 DNN=89.6 CNN=76.4	BNN=37.3 DNN=10.4 CNN=23.6
Qummar et al., 2019 [31]	Ensemble of five deep CNN models (Resnet50, Inceptionv3, Xception, Dense121, Dense169);	80.8	19.2
Bhardwaj et al., 2021 [32]	convolution neural networks (CNNs) with statistical feature selection module and SVM classifier	90.01	9.99
Nazih et al., 2023 [33]	CNN models like ResNet50, InceptionV3, and VGG19.	82.5	17.5
Atwany et al., 2022 [34]	Deep learning methods	91	9
Proposed XAI based model		94	6

posed model gives accurate results as compared to previously published methods. The proposed XAI based model achieves 94% accuracy and 6% miss-rate.

VI. CONCLUSION

There are numerous difficulties associated with diagnosing diabetic retinopathy, such as diverse forms of the disease and initial manifestations that may be easily overlooked, which necessitate highly accurate interpretation and often lead to biases and inconsistencies. Traditional AI models, though they enhance diagnostic precision, typically function as black boxes, providing little insight into their decision-making processes. This lack of transparency makes it difficult for healthcare professionals to trust and effectively use these models. This proposed research addresses this issue by integrating Explainable AI (XAI), transforming the black box into a white box that offers clear and interpretable insights into the model's decisions. The LIME explanations for test images highlight the specific retinal regions that influence predictions, thereby enhancing trust and reliability in clinical practice. The convergence history plots confirm the model's learning progress, showing consistent improvements in both loss reduction and accuracy increment over the training and validation epochs. The proposed model achieves a diagnostic accuracy of 94%, surpassing traditional methods by offering enhanced interpretability and practical utility for clinical use.

VII. LIMITATIONS AND FUTURE WORK

Although the proposed research provides a significant improvement in diagnosing diabetic retinopathy through the development of XAI, there are a few limitations. This research may be limited by a relatively small dataset and the technical addition of XAI may lead to enhancements in computational complexity, which could be challenging in implementing resource-constrained healthcare environments.

In the future, the dataset should be expanded to improve the generalization of the model, and other XAI techniques should also be examined to provide more comprehensive interpretability. Furthermore, the fine-tuning of the model for its real-time applicability in the clinical environment will also be critical.

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